# IGCSE 0607 Practice Questions — E5.5

## Angles

**Total Questions:** 12 — **Total Marks:** 69 Marks

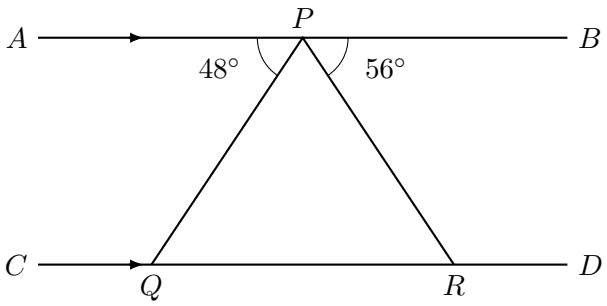
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**1** In the diagram, the line $AB$ is parallel to the line $CD$.
$P$ is a point on $AB$.
$Q$ and $R$ are points on $CD$.
The line $PQ$ and the line $PR$ are drawn.
Angle $APQ = 48°$ and angle $BPR = 56°$.




NOT TO SCALE

**(a)** Write down the value of angle $PQR$.

Angle $PQR =$ ..... ° **[1]**

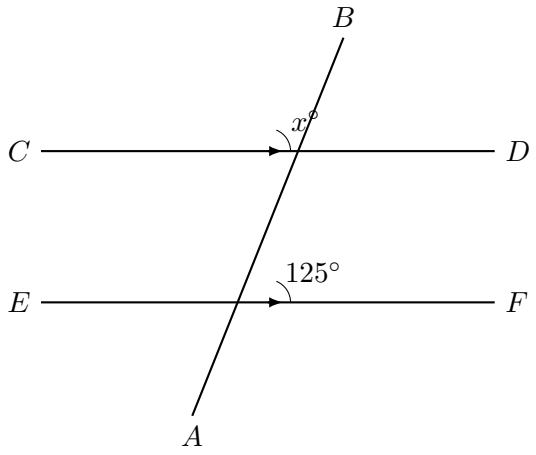
**(b)** Write down the value of angle $PRQ$.

Angle $PRQ =$ ..... ° **[1]**

**(c)** Work out the value of angle $QPR$.

Angle $QPR =$ ..... ° **[1]**

**2** The diagram shows a straight line $AB$ intersecting two parallel lines, $CD$ and $EF$.




**(a)** Find the value of $x$.

$x =$ ..... **[1]**

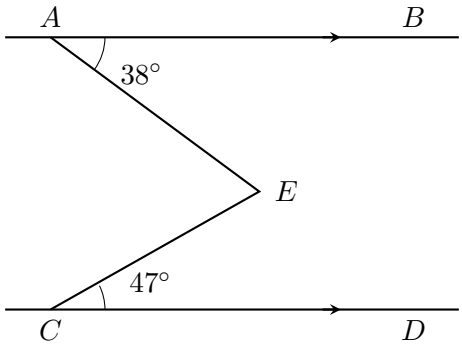
**(b)** Give a geometric reason for your answer to part (a).

..... **[1]**

**(c)** Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

**3** In the diagram, the line $AB$ is parallel to the line $CD$.
The point $E$ lies between the two parallel lines.




NOT TO SCALE

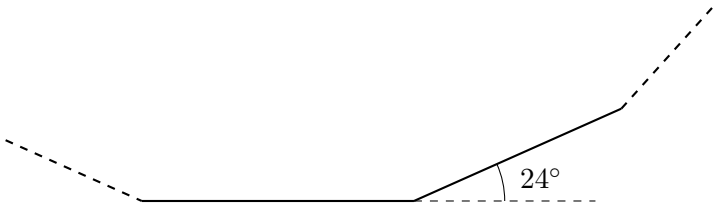
**(a)** Calculate the size of angle $AEC$.

Angle $AEC$ = ..... ° **[2]**

**(b)** A regular polygon has an exterior angle of 24°.
Work out the number of sides of this polygon.

..... **[2]**

**4** The diagram shows part of a regular polygon with $n$ sides.
The exterior angle of this polygon is 24°.




**(a)** Calculate the value of $n$.

$n =$ ..... **[2]**

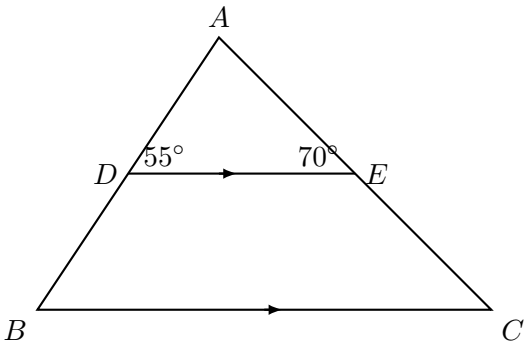
**(b)** Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

**(c)** A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

**5** The diagram shows a triangle $ABC$ with a line $DE$ drawn parallel to $BC$.
$D$ lies on $AB$ and $E$ lies on $AC$.
Angle $ADE = 55°$ and angle $AED = 70°$.




**(a)** Calculate angle $DAE$.

Angle $DAE =$ ..... ° **[2]**

**(b)** Calculate angle $ABC$.

Angle $ABC =$ ..... ° **[1]**

**(c)** Explain why triangle $ADE$ is an isosceles triangle.

**[2]**

**6** Answer the following questions about polygons and angles.

**(a)** An irregular polygon has $n$ sides.
Three of the interior angles are each $140°$.
The remaining $(n - 3)$ interior angles are each $165°$.

Work out the value of $n$.

$n =$ ..... **[4]**

**(b)** In triangle $ABC$, the angles are $\angle A = a°$, $\angle B = b°$ and $\angle C = c°$.
It is given that $c - b = a$.

Prove that $ABC$ is a right-angled triangle.

**[2]**

**(c)** Polygon $P$ is a regular polygon with 12 sides.
Polygon $Q$ is a regular polygon with $k$ sides.
The exterior angle of Polygon $Q$ is half the size of the exterior angle of Polygon $P$.
Polygon $P$ and Polygon $Q$ share a common edge $AB$.

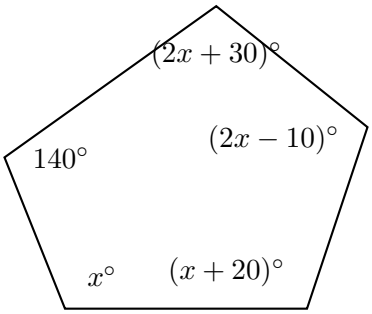
**(i)** Find the value of $k$.

$k =$ ..... **[2]**

**(ii)** Work out the size of the angle at vertex $B$ that lies outside both polygons.

..... ° **[2]**

**7** The diagram shows an irregular pentagon.
The interior angles of the pentagon are $x°$, $(x + 20)°$, $(2x - 10)°$, $(2x + 30)°$ and $140°$.




NOT TO SCALE

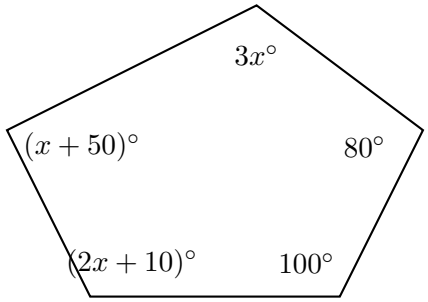
**(a)** Calculate the value of $x$.

$x =$ ..... **[3]**

**(b)** Find the size of the largest interior angle of the pentagon.

..... ° **[2]**

**8** The diagram shows an irregular pentagon.




NOT TO SCALE

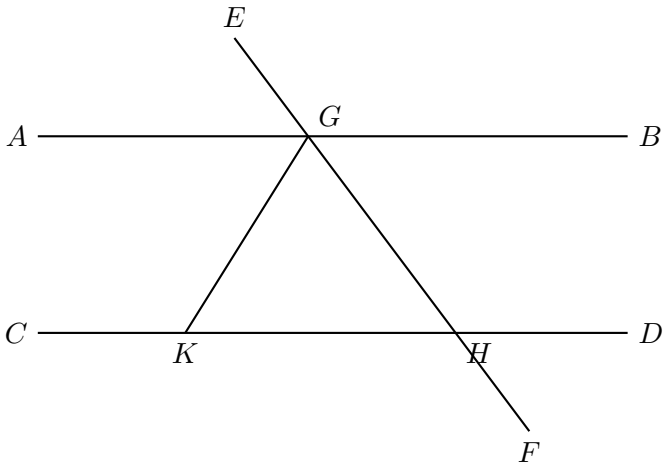
**(a)** Work out the value of $x$.

$x =$ ..... **[3]**

**(b)** Another regular polygon has an interior angle that is 4 times the size of its exterior angle.
Calculate the number of sides of this polygon.

..... **[3]**

**9** In the diagram, the line $AB$ is parallel to the line $CD$.
The transversal $EF$ intersects $AB$ at $G$ and $CD$ at $H$.
The point $K$ lies on $CD$.




**(a)** Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of $x$.
Give a geometrical reason for your working.

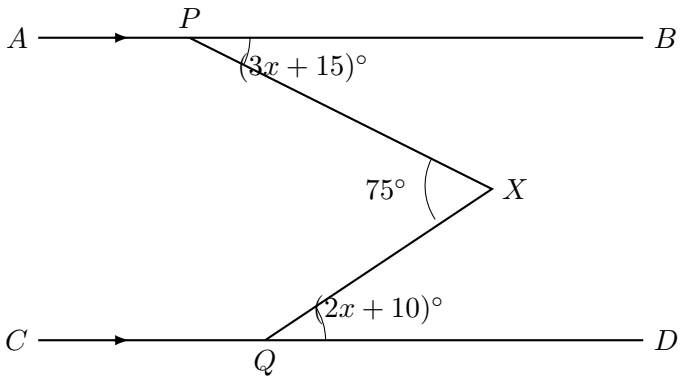
$x =$ ..... , Reason: ..... **[3]**

**(b)** Angle $HGK = 35°$.
Calculate angle $GKH$.

..... ° **[3]**

**10** **(a)** In the diagram, the line $AB$ is parallel to the line $CD$.
$X$ is a point between $AB$ and $CD$.
The point $P$ is on $AB$ and the point $Q$ is on $CD$.
Angle $BPX = (3x + 15)^\circ$, angle $DQX = (2x + 10)^\circ$ and angle $PXQ = 75^\circ$.

Find the value of $x$.




NOT TO SCALE

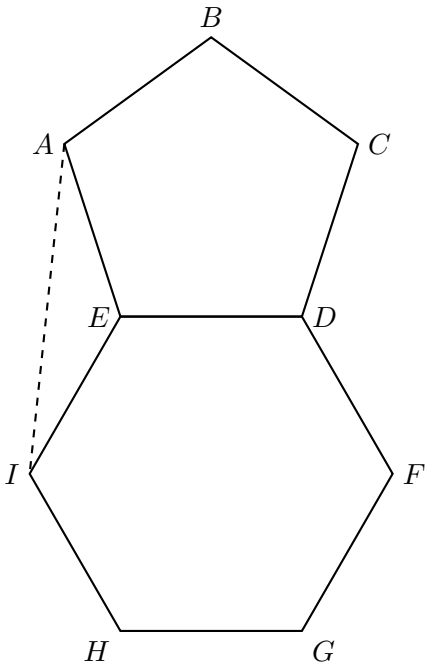
$x =$ ..... **[3]**

**(b)** Polygon $P$ is a regular polygon with $n$ sides.
Polygon $Q$ is a regular polygon with $3n$ sides.
The size of an interior angle of polygon $Q$ is $16^\circ$ greater than the size of an interior angle of polygon $P$.

Calculate the value of $n$.

$n =$ ..... **[4]**

**11** The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge $ED$.




**(a)** Calculate angle $CDE$.

Angle $CDE =$ ..... ° **[2]**

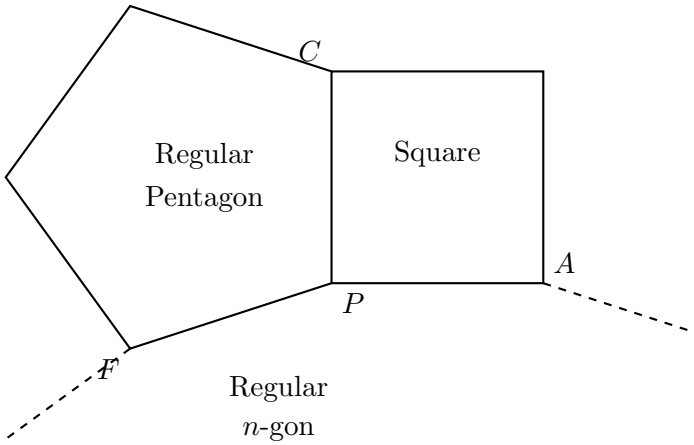
**(b)** Calculate angle $AEI$.

Angle $AEI =$ ..... ° **[2]**

**(c)** Calculate angle $EAI$.

Angle $EAI =$ ..... ° **[3]**

**12** The diagram shows three regular polygons meeting at the point $P$.
The polygons are a square, a regular pentagon, and a regular polygon with $n$ sides.
The square shares the side $PC$ with the pentagon and the side $PA$ with the $n$-gon.
The pentagon shares the side $PF$ with the $n$-gon.
They meet with no gaps and do not overlap.




**(a)** Calculate the interior angle of a regular pentagon.

..... ° **[2]**

**(b)** Find the value of $n$.

$n =$ ..... **[3]**

**(c)** The vertices $A$ and $F$ of the $n$-gon are joined by a straight line.
Calculate angle $PFA$.

..... ° **[2]**